

Unit-1

Random Variables and Random Processes

Definition of Signal:

In communication systems, signal processing and Electrical Engineering, a signal is defined as a function that "conveys information about the behavior or attributes of some phenomenon".

The *IEEE Transactions on Signal Processing* states that the term "signal" includes audio, video, speech, image, communication, geophysical, sonar, radar, medical and musical signals. In a *communication system*, a *transmitter* encodes a *message* to a signal, which is carried to a *receiver* by the *communications channel*.

Continuous-time Signal:

A continuous-time signal is a signal that can be defined at every instant of time. A continuous-time signal contains values for all real numbers along the X-axis. It is denoted by $x(t)$. Figure 1(a) shows continuous-time signal.

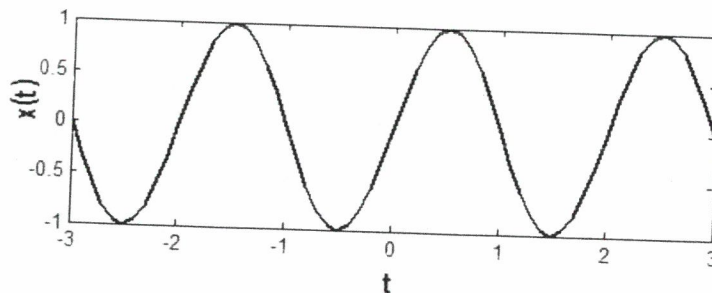


Fig.1.1 Continuous-time signal

Discrete-time Signal:

Signals that can be defined at discrete instant of time is called discrete time signal. Basically, discrete time signals can be obtained by sampling a continuous-time signal. It is denoted as $x(n)$. Figure 1(b) shows discrete-time signal.

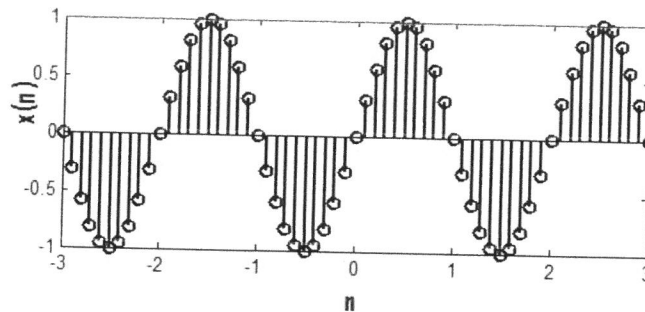


Fig.1.2 Discrete-time signal

Deterministic and Stochastic Signals

A random signal cannot be described by any mathematical function, whereas a deterministic signal is one that can be described mathematically. A common example of random signal is noise. Further Continuous signals can be classified into two groups:

Deterministic Signals:

A signal is said to be deterministic if there is no uncertainty with respect to its value at any instant of time. Or, signals which can be defined exactly by a mathematical formula are known as deterministic signals.

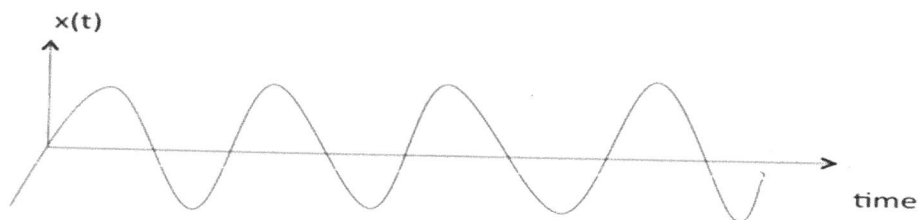


Figure 1.3 Deterministic Signal

- **Stochastic or Non-deterministic Signals:** A signal is said to be non-deterministic if there is uncertainty with respect to its value at some instant of time. Non-deterministic signals are random in nature hence they are called random signals.

Random signals cannot be described by a mathematical equation. They are modeled in probabilistic terms.

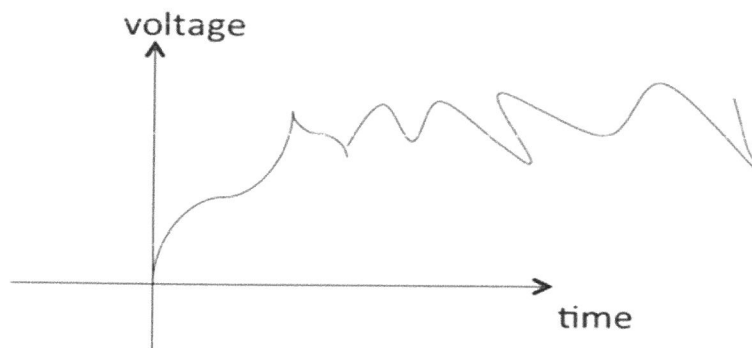


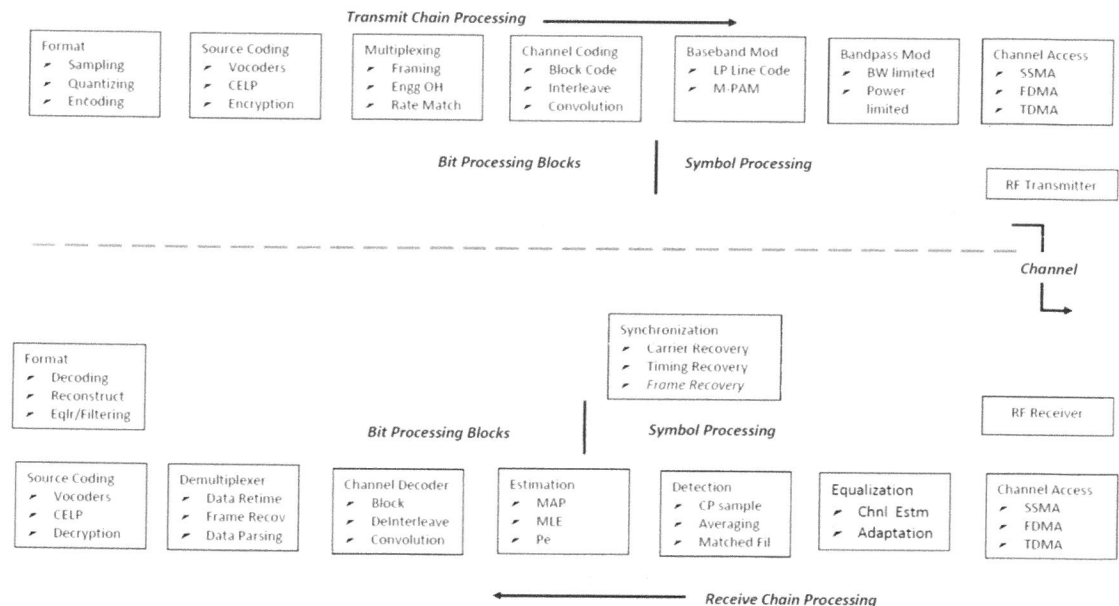
Figure 1.4 Stochastic Signal

Difference between Deterministic and Random Signals

In signal processing, we generally classify signals as deterministic or random. A signal is defined as deterministic if its values at any point in time can be defined precisely by a mathematical equation. For example, the signal $x(n) = \sin(\pi n/4)$ is deterministic. On the other hand, random signals have uncertain values and are usually described using their statistics. A discrete-time random process involves an ensemble of sequences $x(n,m)$ where m is the index of the m -th sequence in the ensemble and n is the time index.

In practice, one does not have access to all possible sample signals of a random process. Therefore, the determination of the statistical structure of a random process is often done from the observed waveform. This approach becomes valid and simplifies random signal analysis if the random process at hand is ergodic. Ergodicity implies that the statistics of a random process can be determined using time-averaging operations on a single observed signal. Ergodicity requires that the statistics of the signal are independent of the time of observation.

Random signals whose statistical structure is independent of time of origin are generally called stationary. More specifically, a random process is said to be wide sense stationary if its statistics, upto the second order, are independent of time. Although it is difficult to show analytically that signals with various statistical distributions are ergodic, it can be shown that a stationary.



Block Diagram of a Communication System

A digital communication system that is required to provide practically error-free communication across a wireless channel. Unfortunately, the wireless channel is subject to *uncertainties*, the sources of which include:

- *noise*, internally generated due to thermal agitation of electrons in the conductors and electronic devices at the front-end of the receiver;
- *fading* of the channel, due to the multipath phenomenon—an inherent characteristic of wireless channels;
- *interference*, representing spurious electromagnetic waves emitted by other communication systems or microwave devices operating in the vicinity of the receiver.

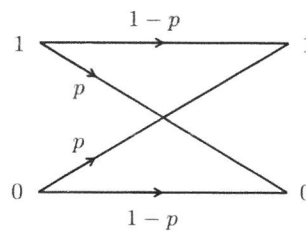
To account for these uncertainties in the design of a wireless communication system, we need a probabilistic model of the wireless channel.

- Apriori Probabilities
- Aposteriori Probabilities
- Likelihood Probabilities
- Baye's Theorem

Example-1

In a binary communication system, the input bits transmitted over the channel are either 0 or 1 with probabilities 0.3 and 0.7, respectively. When a bit is transmitted over the channel, it can be either received correctly or incorrectly (due to channel noise). Let us assume that if a 0 is transmitted, the probability of it being received in error (i.e., being received as 1) is 0.01, and if a 1 is transmitted, the probability of it being received in error (i.e., being received as 0) is 0.1.

1. What is the probability that the output of this channel is 1?
2. Assuming we have observed a 1 at the output of this channel, what is the probability that the input to the channel was a 1?



$$\begin{aligned}
 P(X=0) &= 0.3; & P(X=1) &= 0.7; \\
 P(Y=0|X=0) &= 0.99; & P(Y=1|X=0) &= 0.01; \\
 P(Y=0|X=1) &= 0.1; & P(Y=1|X=1) &= 0.9.
 \end{aligned}$$

1. From the total probability theorem, we have

$$\begin{aligned}
 P(Y=1) &= P(Y=1, X=0) + P(Y=1, X=1) \\
 &= P(X=0)P(Y=1|X=0) + P(X=1)P(Y=1|X=1) \\
 &= 0.3 \times 0.01 + 0.7 \times 0.9 \\
 &= 0.003 + 0.63 \\
 &= 0.633.
 \end{aligned}$$

2. From the Bayes rule, we have

$$\begin{aligned}
 P(X=1|Y=1) &= \frac{P(X=1)P(Y=1|X=1)}{P(X=0)P(Y=1|X=0) + P(X=1)P(Y=1|X=1)} \\
 &= \frac{0.7 \times 0.9}{0.3 \times 0.01 + 0.7 \times 0.9} \\
 &= \frac{0.63}{0.633} \\
 &\approx 0.995.
 \end{aligned}$$

Example-2

$$\mathbb{P}[B|A] = \frac{\mathbb{P}[A|B]\mathbb{P}[B]}{\mathbb{P}[A]}$$

Radar Detection

Radar, a remote sensing system, operates by transmitting a sequence of pulses and has its receiver listen to echoes produced by a target (e.g., aircraft) that could be present in its surveillance area.

Let the events A and B be defined as follows:

$A = \{\text{a target is present in the area under surveillance}\}$

$A^c = \{\text{there is no target in the area}\}$

$B = \{\text{the radar receiver detects a target}\}$

In the radar detection problem, there are three probabilities of particular interest:

$\mathbb{P}[A]$ probability that a target is present in the area; this probability is called the *prior probability*.

$\mathbb{P}[B|A]$ probability that the radar receiver detects a target, given that a target is actually present in the area; this second probability is called the *probability of detection*.

$\mathbb{P}[B|A^c]$ probability that the radar receiver detects a target in the area, given that there is no target in the surveillance area; this third probability is called the *probability of false alarm*.

Suppose these three probabilities have the following values:

$$\mathbb{P}[A] = 0.02$$

$$\mathbb{P}[B|A] = 0.99$$

$$\mathbb{P}[B|A^c] = 0.01$$

The problem is to calculate the conditional probability $\mathbb{P}[A|B]$ which defines the probability that a target is present in the surveillance area given that the radar receiver has made a target detection.

Applying Bayes' rule, we write

$$\begin{aligned}\mathbb{P}[A|B] &= \frac{\mathbb{P}[B|A]\mathbb{P}[A]}{\mathbb{P}[B]} \\ &= \frac{\mathbb{P}[B|A]\mathbb{P}[A]}{\mathbb{P}[B|A]\mathbb{P}[A] + \mathbb{P}[B|A^c]\mathbb{P}[A^c]} \\ &= \frac{0.99 \times 0.02}{0.99 \times 0.02 + 0.01 \times 0.98} \\ &= \frac{0.0198}{0.0296} \\ &\approx 0.69\end{aligned}$$

• Example- 3: Three Coin Problem Analogy of Detection and Estimation

C1= 2 Headed

C2= 2 Tailed

C3= H/T

Apriori Prob: $P(C1)=P(C2)=P(C3)=1/3$

Likelyhood Prob: $P(Z=H/C1)=1$, $P(Z=H/C2)=0$, $P(Z=H/C3)=0.5$

Aposteriori Probabilities: Probability that coin is C_i having observed Z ($i=1,2,3$)

Baye's Theorem :

$$P(C_i/Z)=P(Z/C_i)P(C_i)/P(Z)$$

If observation is head ($Z=H$), From which coin it has come from?.

Solution :

- $P(C2/Z=H)= 0$
- $P(C1/Z=H)= 2/3$
- $P(C3/Z=H)= 1/3$

Ensemble Average and Time Average

- Analog Circuits Lab Example
 - Voltage Regulator Experiment
 - Noise Across a Resistor

Random Variables and Random processes

The Theory of probability and random processes is used in many Engineering applications.

Examples include

- 1) Speech recognition System
- 2) Radar System
- 3) A Communication Network

Speech recognition system - The waveform templates may vary considerably from speaker to speaker. This variability may be described by the Theory of probability and random processes, which in turn may be used to develop models for speech production and recognition.

Radar system - Signal detection and Estimation.

The reflected signal from the aircraft is processed to gather information about the aircraft. In addition to being corrupted by a random noise and interference process, the returning signal itself may exhibit randomness as well. From the time of arrival of the reflected signal, we can estimate the distance of the aircraft from the radar site.

Communication Network - The arrival of packets to a node will be random, the node will usually have some buffering capability, allowing the node to temporarily store packets

which it cannot forward immediately. Given a random model of arrival process of packets at a node, the theory of probability and random processes will allow the network designer to determine how large a buffer is needed to ensure a minimal probability of buffer overflow. Other random quantities such as the delay a packet encounters at the node can also be statistically characterized. Congestion throughout the network tends to be very dynamic and so the routing decision must be made using probability.

Introduction to Probability Theory

The theory of probability and random processes is useful for modeling the uncertainty of various events. Performance of many systems is adversely affected by noise, which may often be present in the form of an undesired signal that degrades the performance of the system. Thus, it becomes necessary to design systems that can discriminate against noise and enhance the desired signal.

Stochastic signals have an element of randomness which prevent them from being perfectly reconstructed from a past history. The theory of probability provides a tool to model and analyze phenomena that occur in many diverse fields, such as communications, signal processing, control and computers.

Experiment — An experiment is a procedure we perform that produces some result.

Outcome — An outcome is a possible result of an experiment.

Event — An event is a certain set of outcomes of an experiment.

Sample Space — The collection or set of "all possible" distinct (collectively exhaustive & mutually exclusive) outcomes of an experiment.

Examples

1) Flipping a fair coin once - Discrete Sample space and it is countably finite. Sample space $\Omega = \{H, T\}$

2) Rolling a die once.

$$\Omega = \{1, 2, \dots, 6\}, \text{ countable and finite.}$$

3) Rolling of two dice.

$$\Omega = \{(1,1), (1,2), \dots, (6,6)\}$$

Discrete Sample space, Countably finite.

4) Flip a coin until a tail occurs.

The possible outcomes that may occur are

$$\Omega_1 = \{T\}, \Omega_2 = \{H, T\}, \Omega_3 = \{H, H, T\} \dots$$

In this case the ^{discrete} sample space is countably infinite.

5) Random number generator which selects a number in an arbitrary manner from the semi-closed interval $[0, 1)$. The sample space consists of all real numbers, x for which $0 \leq x < 1$.

$$A = \{x < 1/2\}, B = \{|x - 1/2| < 1/4\},$$

$$C = \{x = 1/2\}$$

Sample space is Continuous and Countably Infinite.

* There are also infinite sets that are uncountable and that are not continuous. (Discrete & Uncountable)

Axioms of probability

- 1) For any event A , $P(A) \geq 0$
- 2) If S is the sample space for a given experiment, $P(S) = 1$
probabilities are normalized so that the maximum value is unity.

- 3) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$

A and B are mutually exclusive / disjoint events.

- 4) For an infinite number of mutually exclusive sets A_i , $i = 1, 2, 3, \dots$, $A_i \cap A_j = \phi$ for all $i \neq j$.

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Theorems -

- 1) For any sets A and B (not necessarily mutually exclusive)
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- 2) $P(\bar{A}) = 1 - P(A)$

- 3) If $A \subset B$, then $P(A) \leq P(B)$

Joint and Conditional probabilities

* If A and B are mutually exclusive, then their joint probability is zero. $P(A, B) = P(\emptyset) = 0$, An impossible event should never happen.

* When A and B are not necessarily mutually exclusive, the joint probability of A and B is,

$$P(A, B) = \lim_{N \rightarrow \infty} \frac{n_{A, B}}{N}$$

where $n_{A, B}$ be the number of times that A and B simultaneously occur in N trials.

$N \rightarrow$ Number of trials.

* The probability of event A depends on the occurrence of event C, is referred to as the conditional probability of A given C. $P(A|C) \rightarrow$ prob of the event A given that the event C has occurred.

$$\text{Ex- } P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B, A)}{P(A)}$$

$$P(A, B) = P(B|A) P(A) = P(A|B) P(B)$$

$$P(A, B, C) = P(C|A, B) P(A, B) = P(C|A, B) P(B|A) P(A)$$

Theorem of total probability

Let $B_1, B_2, \dots, B_n$ be a set of mutually exclusive & exhaustive Events.

that is $B_i \cap B_j = \phi$ for all $i \neq j$ and

$$\bigcup_{i=1}^N B_i = S \Rightarrow \sum_{i=1}^N \Pr(B_i) = 1$$

$$\text{Then } P(A) = \sum_{i=1}^N P(A|B_i) P(B_i)$$

Bayes's Theorem - Let $B_1, B_2, \dots, B_n$ be a set of mutually exclusive and exhaustive Events. Then,

$$P(B_i|A) = \frac{P(A|B_i) P(B_i)}{\sum_{i=1}^N P(A|B_i) P(B_i)}$$

$P(B_i)$ $\rightarrow$ A priori probability of Event B_i

$P(B_i|A)$ $\rightarrow$ A posteriori prob of event B_i given A .

Independence - Two Events A and B are independent

if $P(A|B) = P(A)$. That is the a priori probability of event A is identical to the a posteriori probability of A given B .

$$\text{If } P(A|B) = P(A), \text{ then } P(B|A) = P(B)$$

Then $P(A, B) = P(A) P(B) \rightarrow$ Two Events are statistically independent.

$$\text{If } P(A|B) = P(A) P(B)$$

$$P(A, C) = P(A) P(C)$$

$$P(B, C) = P(B) P(C)$$

$$\text{Then, } P(A, B, C) = P(A) P(B) P(C)$$

Discrete Random Variables

Probability mass function (PMF) $P_X(x)$ of a random variable X is a function that assigns a probability to each possible value of the random variable X .

$$\text{i.e. } P_X(x) = P(X=x)$$

- 1) Bernoulli Random variable, mean = p , $\sigma^2 = p(1-p)$
- 2) Binomial random variable, mean = np , $\sigma^2 = np(1-p)$
- 3) Poisson Random variable, mean = λ , $\sigma^2 = \lambda$
- 4) Geometric random variable, mean = $\frac{p}{1-p}$, $\sigma^2 = \frac{p}{(1-p)^2}$

PMF is used to describe the discrete Random variables only.

PMF is not suitable for Continuous random variables.

PDF (Probability density function) is used to describe the Continuous random variables.

CDF (Cumulative distribution function) is used to describe the Continuous & discrete random variables.

The Cumulative Distribution Function (CDF)

Since the Continuous random variable will typically have a zero probability of taking on a specific value, we avoid talking about such probabilities. Instead, events of the form $\{X \leq x\}$ can be considered.

The CDF of a random variable, X is

$$F_X(x) = P(X \leq x)$$

$$0 \leq F_X \leq 1$$

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

CDF is a monotonic non-decreasing function.

$$\text{for } x_1 < x_2, P(x_1 \leq X \leq x_2) = F_X(x_2) - F_X(x_1)$$

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x \leq 1 \\ 1, & x > 1 \end{cases}$$

If the random variable is discrete, the CDF is discontinuous and has jumps at the specific values that the random variable could take on. Whereas, for Continuous random variable, the CDF is a continuous function.

Discrete random variable $\rightarrow$ Discontinuous CDF (Staircase type)

Continuous random variable $\rightarrow$ Continuous CDF

The Probability Density Function

It is often cumbersome to work with CDF's. An alternative and often more convenient description known as probability density function is often used.

The PDF of the random variable x evaluated at the point x is

$$f_x(x) = \lim_{\epsilon \rightarrow 0} \frac{P(x \leq x < x + \epsilon)}{\epsilon}$$

As the name implies, the PDF is the probability that the random variable X lies in an infinitesimal interval about the point $x = x$, normalized by the length of the interval.

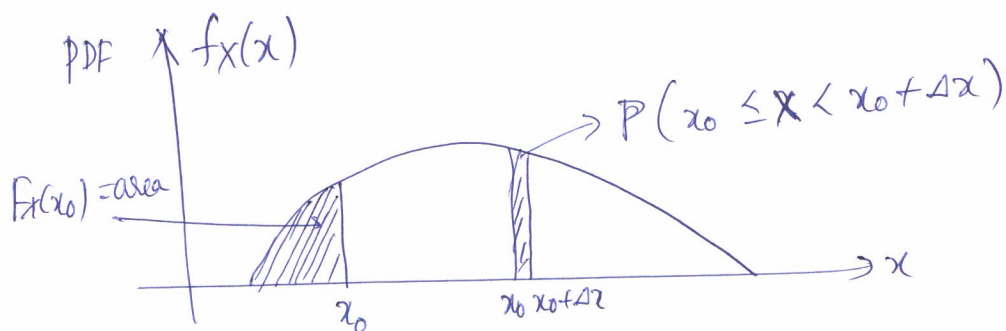
For continuous random variables,

$$P(x \leq x < x + \epsilon) = F_x(x + \epsilon) - F_x(x)$$

So that

$$f_x(x) = \lim_{\epsilon \rightarrow 0} \frac{F_x(x + \epsilon) - F_x(x)}{\epsilon} = \frac{dF_x(x)}{dx}$$

The PDF of a random variable is the derivative of CDF. Conversely, the CDF of a random variable can be expressed as the integral of its PDF.



$$F_x(x) = \int_{-\infty}^x f_x(x) dx$$

* The PDF is a nonnegative function although it is not restricted to be less than unity as with the CDF.

* Properties of PDFs are

$$1) f_X(x) \geq 0$$

$$2) F_X(x) = \int_{-\infty}^x f_X(y) dy$$

$$3) f_X(x) = \frac{dF_X(x)}{dx}$$

$$4) \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$5) \int_a^b f_X(x) dx = P(a < X \leq b)$$

The Gaussian Random Variable

The Gaussian random variable is the most commonly used and of most importance. Many physical phenomenon can be modeled as Gaussian random variables, including the thermal noise encountered in electronic circuits.

PDF of Gaussian random variable is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

$m \rightarrow \text{mean}$, $\sigma^2 \rightarrow \text{variance}$

The special case is when $m=0$ and $\sigma=1$ it is called a standard normal random variable.

CDF of gaussian random variable is

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy \rightarrow \textcircled{1}$$

Operations on a single random variable

Expected value of a random variable — The expected value of a random variable X which has a PDF, $f_X(x)$ is

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$\mu_X = E[X] = \text{Average / mean / expectation / first moment}$$

For discrete random variables,

$$E[X] = \sum_k x_i P_X(x_i)$$

Given a random variable X with PDF $f_X(x)$, the expected value of a function, $g(x)$, of that random variable is given by

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$E[g(x)] = \sum_k g(x_k) P_X(x_k)$$

For any constants a and b ,

$$E[aX + b] = aE[X] + b$$

$$E\left[\sum_{k=1}^N g_k(x)\right] = \sum_{k=1}^N E[g_k(x)]$$

Mean of Sum of functions = Sum of their individual means.

Moments

The n th moment of a random variable X is defined as

$$E[X^n] = \int x^n f_X(x) dx$$

$$E[X^n] = \sum_k (x_k)^n P_X(x_k)$$

Mean, average, expected value $\rightarrow$ first moment $\mu_X = \bar{X} = E[X]$

Variance $\rightarrow$ second moment $\sigma_X^2 = E[(X - \mu_X)^2]$

Skewness $\rightarrow$ third moment $C_3 = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^3\right]$

Kurtosis $\rightarrow$ fourth moment $C_4 = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^4\right]$

Pair of random variables

The joint PDF $f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$

The joint CDF $F_{X,Y}(x,y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u,v) du dv$

Properties

$$1) f_{X,Y}(x,y) \geq 0, \quad P(x_1 < X < x_2, y_1 < Y < y_2) = \int_{y_1}^{y_2} \int_{x_1}^{x_2} f_{X,Y}(x,y) dx dy$$

$$2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1$$

$$3) F_{X,Y}(x,y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u,v) du dv$$

$$4) f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

$$5) f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

* Conditional probability

$$P_{X|Y}(x|y) = P_{X,Y}(x,y) / P_Y(y)$$

* Conditional PDF of a random variable X given that $Y=y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

* The Correlation b/w two random variables is defined as

$$R_{X,Y} = E[XY] = \int \int xy f_{X,Y}(x,y) dx dy$$

Two random variables which have a correlation of zero are said to be orthogonal.

* The Covariance b/w two random variables is

$$\begin{aligned} \text{Cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= \int \int (x-\mu_X)(y-\mu_Y) f_{X,Y}(x,y) dx dy \end{aligned}$$

If the two random variables have a Covariance of zero, then they are said to be uncorrelated.

$$\text{Cov}(X,Y) = R_{X,Y} - \mu_X \mu_Y$$

$$\text{Var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2 \text{Cov}(X,Y)$$

* The Correlation Coefficient of two random variables X and Y

$$\begin{aligned} \text{is } \rho_{XY} &= \frac{\text{Cov}(X,Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y} \\ &= \frac{\sigma_{XY}}{\rho_X \rho_Y} \end{aligned}$$

$$-1 \leq \rho_{XY} \leq 1$$

* If $\rho_{XY} = 0$, random variables are uncorrelated.

$\rho_{XY} = 1$ or -1 implies that X and Y are completely correlated.

* Independence implies uncorrelatedness

* uncorrelatedness does not imply independence

Independent random Variables

* The two events A and B are statistically independent if

$$P(A, B) = P(A) \cdot P(B)$$

$$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y)$$

$$F_{XY}(x, y) = F_X(x) F_Y(y)$$

* Two random variables are statistically independent if their joint CDF factors into a product of the marginal CDF's.

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

$$f_{X|Y}(x|y) = f_X(x) \quad \text{or} \quad f_{Y|X}(y|x) = f_Y(y)$$

Independent and Identically distributed RV's

In many applications, we are able to observe an experiment repeatedly. Each new observation can occur with an independent realization of whatever random phenomena control the experiment. This sort of situation gives rise to independent and identically distributed random variables.

* A sequence of random variables $x_1, x_2, \dots, x_n$ is IID if

$$F_{x_i}(x) = F_x(x) \quad \forall i = 1, 2, \dots, n \text{ (Identically distributed)}$$

and

$$F_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n F_{x_i}(x_i) \text{ (Independent)}$$

* For continuous random variables, the CDF's can be replaced with PDF's

$$p_{x_i}(x) = p_x(x), \quad \forall i = 1, 2, \dots, n$$

$$\text{and } p_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p_{x_i}(x_i)$$

Random Processes

1

A random variable that is a function of time is called random processes. Thus a random processes is a Collection of an infinite number of random variables. Communication signals as well as noises, typically random and varying with time are well characterized by random processes.

Characterization of a Random Process

Consider for instance, a random process described by

$x(t) = A \cos(\omega t + \theta)$, where θ is a random variable uniformly distributed over the range $(0, 2\pi)$. This expression completely describes a random process. It is not always possible to be able to describe a random process analytically.

Without a specific model, we may have just an ensemble obtained experimentally. The ensemble has the complete information about the random process.

The complete information of several dependent RV's is provided by the joint PDF of those variables. The 'n' RV's $x_1, x_2, x_3, \dots, x_n$ generated by the amplitudes at $t = t_1, t_2, \dots, t_n$ respectively are dependent in general. For the n samples, they are fully characterized by the n-th order joint probability density function or the nth order joint CDF,

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = P[X(t) \leq x_1; X(t) \leq x_2; \dots, X(t_n) \leq x_n] \quad (2)$$

Joint PDF is

$$P_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = \frac{\partial^n}{\partial x_1 \partial x_2 \dots \partial x_n} F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) \rightarrow (1)$$

It can be shown that the random process is completely described by the n -th order joint PDF for all n (upto ∞) and for any choice of $t_1, t_2, \dots, t_n$.

We can always derive a lower order PDF from higher order PDF by simple integration.

$$P_X(x_1; t_1) = \int_{-\infty}^{\infty} P_X(x_1, x_2; t_1, t_2) dx_2$$

Hence, when the n th order PDF is available, there is no need to specify PDF's of order lower than n .

The mean μ_X of a random process $x(t)$ can be determined from the first-order PDF as

$$E[X(t)] = \int_{-\infty}^{\infty} x P_X(x, t) dx$$

which is typically a deterministic function of time t .

Why do we need Ensemble Statistics

(3)

To specify a random process, we need ensemble statistics. For instance, to determine the PDF $P_X(x_1)$, we need to find the values of all the sample functions at $t = t_1$. This is ensemble statistics.

* In deterministic signals, we are used to studying the data of a waveform as a function of time.

To understand the necessity of ensemble statistics, consider the problem of threshold detection (Ex:- $1 \rightarrow p(t)$, $0 \rightarrow -p(t)$)

$A_p \rightarrow$ peak amplitude.

1 is transmitted, the received sample value is $A_p + n$

0 is transmitted, the received sample value is $-A_p + n$

To find the error probability, we repeat the experiment N times and see how many times the noise at $t = t_s$ is less than $-A_p$.

For any statistical specification & characterization of the random process, we need to average over the entire ensemble.

Autocorrelation Function of a Random process

The Spectral Content of a process depends on the rapidity of the amplitude change with time. This can be measured by correlating amplitudes at t_1 and $t_1 + \tau$. We can use Correlation to measure the similarity of amplitudes at t_1 and $t_2 = t_1 + \tau$.

If the Rvs $x(t_1)$ and $x(t_2)$ are denoted by x_1 and x_2 respectively, then for a real random process, the autocorrelation function $R_x(t_1, t_2)$ is defined as

$$R_x(t_1, t_2) = E [x(t_1) x(t_2)] \rightarrow \textcircled{2}$$

Eqn $\textcircled{2}$ indicates the similarity b/w Rvs $x(t_1)$ and $x(t_2)$.

It is computed by multiplying amplitudes at t_1 and t_2 of a sample function and then averaging this product over the ensemble.

It can be seen that for a small τ , the product $x_1 x_2$ will be positive for most sample functions of $x(t)$, but the product $y_1 y_2$ is equally likely to be positive or negative.

x_1 and x_2 will show correlation for considerably larger values of τ , whereas y_1 and y_2 will loose correlation quickly, even for small τ . Thus $R_x(t_1, t_2)$, the autocorrelation function of $x(t)$ provides valuable information about the frequency content of the process.

$$R_x(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 P_X(x_1, x_2; t_1, t_2) dx_1 dx_2$$

Properties of the Autocorrelation Function

④

1. The autocorrelation function evaluated at $\tau=0$, $R_{xx}(0)$, is the average normalized power in the random process $x(t)$.

$$R_{xx}(0) = E[x^2(t)]$$

2. Autocorrelation function of a WSS random process is an even function. i.e., $R_{xx}(\tau) = R_{xx}(-\tau)$

Any function of τ which is not even cannot be a valid autocorrelation function.

3. The autocorrelation function of a WSS random process is maximum at the origin. i.e. $|R_{xx}(\tau)| \leq R_{xx}(0)$ for all τ .

4. If $x(t)$ is ergodic and has no periodic components, then

$$\lim_{\tau \rightarrow \infty} R_{xx}(\tau) = \mu_x^2 = \{E(x(t))\}^2$$

5. If $x(t)$ has a periodic component, then $R_{xx}(\tau)$ will have a periodic component with same period.

(2)

* the autocovariance function $C_{xx}(t_1, t_2)$ of a continuous time random process, $x(t)$ is defined as the covariance of $x(t_1)$ and $x(t_2)$

$$C_{xx}(t_1, t_2) = \text{Cov}(x(t_1), x(t_2)) = E[(x(t_1) - \mu_{x(t_1)}) \cdot (x(t_2) - \mu_{x(t_2)})]$$

Autocovariance function for random variables, can be used to write for random process as in terms of autocorrelation function & mean function.

$$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2) - \mu_{x(t_1)} \mu_{x(t_2)}$$

The autocovariance function is helpful when studying random processes which can be represented as the sum of a deterministic signal $s(t)$ + zero mean noise process $N(t)$.

If $x(t) = s(t) + N(t)$, then

$$R_{xx}(t_1, t_2) = E[(s(t_1) + N(t_1))(s(t_2) + N(t_2))]$$
$$= s(t_1) s(t_2) + R_{NN}(t_1, t_2)$$

↑
deterministic signal

($\because$ Using the fact that $\mu_N(t) = 0$)

If the signal is strong compared to the noise, the deterministic part will dominate the autocorrelation function, and thus $R_{xx}(t_1, t_2)$ will not tell us much about the randomness in the process $x(t)$.

On the other hand, the autocovariance function is

$$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2) - s(t_1) s(t_2)$$
$$= R_{NN}(t_1, t_2) = C_{NN}(t_1, t_2)$$

$\therefore$ the autocovariance function allows us to isolate the noise which is the source of randomness in the process.

* For a pair of random process $x(t)$ and $y(t)$, the cross-correlation function is defined as

$$R_{xy}(t_1, t_2) = E[X(t_1) Y(t_2)]$$

The cross-covariance function is

$$\begin{aligned} C_{xy}(t_1, t_2) &= E[(X(t_1) - \mu_x(t_1)) \cdot (Y(t_2) - \mu_y(t_2))] \\ &= R_{xy}(t_1, t_2) - \mu_x(t_1) \mu_y(t_2) \end{aligned}$$

Power Spectral density (PSD)

PSD $S_X(f)$ of a random process $x(t)$ is the ensemble average of the PSD's of all sample functions.

$$S_X(f) = \lim_{T \rightarrow \infty} \left[\frac{|X_T(f)|^2}{T} \right] \quad \text{W/Hz}$$

where $X_T(f) \rightarrow$ Fourier transform of the time-truncated random process.

$$X_T(t) = x(t) \Pi(t/T)$$

PSD is the Fourier transform of the autocorrelation function $R_X(\tau)$ of the process $x(t)$ is

$$R_X(\tau) \longleftrightarrow S_X(f)$$

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau$$

The autocorrelation function $R_X(\tau)$ for real processes is an even function of τ .

$S_X(f)$ is also an even function of f .

The mean square value $\overline{x^2(t)}$ of the random process $x(t)$ is $R_X(0)$,

$$R_X(0) = \overline{x(t) x(t)} = \overline{x^2(t)} = \overline{x^2}$$

↑
Ensemble average of the squares of all sample function amplitudes at any instant t .

Classification of Random processes

(5)

1. Stationary and NonStationary Random processes :

A random process whose statistical characteristics do not change with time is classified as a stationary random process.

The PDF of x at t_1 and $t_2 = t_1 + t_0$ must be same,

$P_x(x, t_1)$ and $P_x(x, t_2)$ must be identical for a stationary random process. This is possible only if $P_x(x, t)$ is independent of 't'.

First order density of a stationary random process can be expressed as

$$P_x(x, t) = P_x(x)$$

For a real stationary process,

$$R_x(t_1, t_2) = R_x(t_2 - t_1)$$

$$R_x(\tau) = E[x(t) x(t + \tau)]$$

For a stationary process, the joint PDF for x_1 and x_2 must also depend only on $t_2 - t_1$.

Similarly, higher order PDFs are all independent of the choice of origin. That is,

$$\begin{aligned} P_x(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) &= P_x(x_1, x_2, \dots, x_n; t_1 - t, t_2 - t, \dots, t_n - t) \\ &= P_x(x_1, x_2, \dots, x_n; 0, t_2 - t_1, \dots, t_n - t_1) \end{aligned}$$

- (6)
- * The random process $x(t)$ representing the temperature of a city is an example of a nonstationary random process because the temperature statistics (mean) depend on the time of the day.
 - * Noise process is stationary because its statistics (mean, mean square value) do not change with time.

2. Wide-sense (weakly) Stationary processes

A process that is not stationary in the strict sense, may yet have a mean value and an autocorrelation function that are independent of the shift of time or origin.

$$E[X(t)] = \text{Constant}$$

$$R_x(t_1, t_2) = R_x(\tau), \quad \tau = t_2 - t_1$$

Such a process is known as a wide-sense stationary or weakly stationary process.

Stationary is a stronger condition than wide-sense stationary.

No truly stationary process can occur in real life. All processes in practice are nonstationary because they begin at some finite time and terminate at some finite time.

The stationary assumption allows a manageable mathematical model.

Ergodic wide-sense stationary processes

(8)

$\bar{x}(t) \rightarrow$ ensemble average of sample function amplitudes at t

$R_x(t_1, t_2) \rightarrow$ ensemble average of the product of sample function amplitudes $x(t_1)$ and $x(t_2)$.

We can also define time averages for each sample function.

time mean $\bar{x}(t)$ of a sample function $x(t)$ is

$$\tilde{x}(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$$

the time autocorrelation function $R_x(z)$ is

$$R_x(z) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t+z) dt$$

For Ergodic (wide-sense) stationary processes, ensemble averages are equal to the time averages of any sample function. For an ergodic process $x(t)$,

$$E[x(t)] = \tilde{x}(t) = \bar{x}(t)$$

$$R_x(z) = R_x(z)$$

Here are the two averages for ergodic wide-sense stationary processes.

In practice many of the stationary processes are ergodic with respect to at least low-order statistics such as mean & autocorrelation.

Ergodic process

A random process is said to be ergodic if time averages are equal to Ensemble averages.

* Time averages are taken along the time.

Ex - ① If the time averages are taken from $-T$ to T , then total time is $2T$. ($T \rightarrow \infty$).

$$\mu_x(T) = \frac{1}{2T} \int_{-T}^T x(t) dt \rightarrow \textcircled{1}$$

② If time averages are taken from $-T/2$ to $T/2$ the total time is T ($T \rightarrow \infty$)

$$\mu_x(T) = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \rightarrow \textcircled{2}$$

Equation ① and ②, result is same.

* Ensemble averages are taken at some fixed instants.
Ensemble averages are taken across the process.

A random process is said to be ergodic in mean if Ensemble mean equals time mean.

Ex: $x(t) = \text{random process} = A \cos(2\pi f_c t + \phi)$

$\phi \rightarrow \text{random variable} \rightarrow \text{uniformly distributed b/w } 0 \text{ to } 2\pi$

$$f_{\phi(t)}(\phi) = \begin{cases} \frac{1}{2\pi} & 0 < \phi < 2\pi \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \mu_x(t) &= E[x(t)] = \int_{-\infty}^{\infty} x(t) f_{x(t)}(x) dx \\ &= \int_0^{2\pi} A \cos(2\pi f_c t + \phi) \cdot \frac{1}{2\pi} d\phi \\ &= \frac{A}{2\pi} \int_0^{2\pi} \cos(2\pi f_c t + \phi) d\phi \\ &= \frac{A}{2\pi} \left[\sin(2\pi f_c t + \phi) \right]_0^{2\pi} = 0 = \text{Ensemble mean} \end{aligned}$$

Time mean :-

$$\begin{aligned} \mu_x(t) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} A \cos(2\pi f_c t + \phi) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left(\frac{A}{2\pi f_c} \right) \left[\sin(2\pi f_c t + \phi) \right]_{-T/2}^{T/2} \\ &= \lim_{T \rightarrow \infty} \frac{A}{2\pi f_c T} \left[\sin(2\pi f_c T/2 + \phi) - \sin(2\pi f_c (-T/2) + \phi) \right] \\ &= \frac{A}{2\pi} \left[\sin(\pi + \phi) - \sin(-\pi + \phi) \right] \quad \left(\because \frac{1}{T} = f_c \right) \\ &= 0 \end{aligned}$$

Power Spectral density of WSS process

①

The term filter may represent the channel of a communication system.

$H(f) \rightarrow$ frequency response of the filter

$$h(z_1) = \int_{-\infty}^{\infty} H(f) \exp(j2\pi f z_1) df \rightarrow (1a)$$

Substitute (1a) into equation (3), we get

$$E[Y^2(t)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} H(f) \exp(j2\pi f z_1) df \right] h(z_2) R_{xx}(z_1 - z_2) dz_1 dz_2$$

$$= \int_{-\infty}^{\infty} \left[H(f) \int_{-\infty}^{\infty} dz_2 h(z_2) \int_{-\infty}^{\infty} R_{xx}(z_1 - z_2) \exp(j2\pi f z_1) dz_1 \right] df$$

Let $\tau = z_1 - z_2$, then the above expression becomes,

$$E[Y^2(t)] = \int_{-\infty}^{\infty} H(f) \left[\int_{-\infty}^{\infty} h(z_2) \exp(j2\pi f z_2) dz_2 \cdot \int_{-\infty}^{\infty} R_{xx}(\tau) \exp(-j2\pi f \tau) d\tau \right] df$$

$\int_{-\infty}^{\infty} h(z_2) \exp(j2\pi f z_2) dz_2 = H^*(f) =$ the complex conjugate of the frequency response of the filter.

$$E[Y^2(t)] = \int_{-\infty}^{\infty} H(f) \cdot H^*(f) \int_{-\infty}^{\infty} R_{xx}(\tau) \exp(-j2\pi f \tau) d\tau \cdot df$$

$$E[Y^2(t)] = \int_{-\infty}^{\infty} |H(f)|^2 \left[\int_{-\infty}^{\infty} R_{xx}(\tau) \exp(-j2\pi f \tau) d\tau \right] df \rightarrow (2a)$$

In equation (2a), the term inside the brackets is the fourier transform of the autocorrelation function $R_{xx}(\tau)$ of the input process $x(t)$. (2)

we define a new function

$$S_{xx}(f) = \int_{-\infty}^{\infty} R_{xx}(\tau) \exp(-j2\pi f \tau) d\tau$$

The new function $S_{xx}(f)$ is called power spectral density or power spectrum of the WSS process $x(t)$.

Thus we have

$$R[Y^2(t)] = \int_{-\infty}^{\infty} |H(f)|^2 S_{xx}(f) df \rightarrow (3a)$$

Eqn (3a) can be stated as,

The mean square value of the output of a stable linear time invariant filter in response to a weakly stationary process is equal to the integral over all frequencies of the power spectral density of the input process multiplied by the squared magnitude response of the filter.

Wiener - Khintchine relations

3

$R_{xx}(z)$ and $S_{xx}(f)$ form a Fourier-transform pair.

$$S_{xx}(f) = \int_{-\infty}^{\infty} R_{xx}(z) \exp(-j2\pi f z) dz$$

$$R_{xx}(z) = \int_{-\infty}^{\infty} S_{xx}(f) \exp(j2\pi f z) df$$

For practical application purposes, the power spectral density is the more useful function of the two relations above.

Power of a Random process

The power P_X (Avg power) of a wide-sense random process $x(t)$ is its mean square value $\overline{x^2}$.

$$R_X(z) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f z} df$$

$$P_X = \overline{x^2} = R_X(0) = \int_{-\infty}^{\infty} S_X(f) df$$

Because $S_X(f)$ is an even function of f , we have

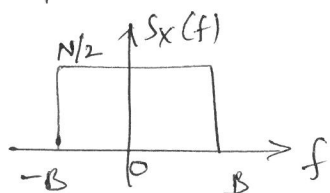
$$P_X = \overline{x^2} = 2 \int_0^{\infty} S_X(f) df$$

$P_X \rightarrow$ the area under the PSD.

$P_X = \overline{x^2} \rightarrow$ the Ensemble mean of the square amplitudes of the sample functions at any instant.

* the PSD may not exist for processes that are not wide-sense stationary.

Ex - Determine the autocorrelation function $R_X(z)$ and the power P_X of a low-pass random process with a white noise PSD $S_X(f) = N/2$



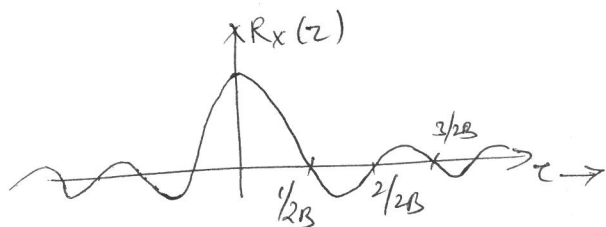
Soln :-

$$S_X(f) = \frac{N}{2} \Pi\left(\frac{f}{2B}\right)$$

$$R_X(z) = NB \operatorname{Sinc}(2\pi Bz)$$

$$P_X = \overline{x^2} = R_X(0) = NB$$

$$\text{or } P_X = 2 \int_0^{\infty} S_X(f) df = 2 \int_0^B \frac{N}{2} df = NB$$



Gaussian Random Processes

(5)

A random process $x(t)$ for which any n samples, $x_1 = x(t_1)$, $x_2 = x(t_2)$, $\dots$ $x_n = x(t_n)$, taken at arbitrary points in time $t_1, t_2 \dots t_n$ from a set of jointly gaussian random variables for any $n = 1, 2, 3 \dots$ is a Gaussian random process.

In a vector notation, the vector of n samples

$X = [x_1, x_2, \dots, x_n]^T$, will have a joint PDF given by

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n |C_{XX}|}} \exp\left(-\frac{1}{2}(x - \mu_X)^T \cdot C_{XX}^{-1} \cdot (x - \mu_X)\right)$$

Mean and autocorrelation functions provide sufficient information to specify the joint PDF for any number of samples of a Gaussian random process.

Note that since any n th order PDF is completely specified by $\mu_X(t)$ and $R_{XX}(t_1, t_2)$, if a gaussian random process is WSS, then the mean and autocorrelation functions will be invariant to a time shift and therefore any PDF will be invariant to a time shift. Hence, any WSS Gaussian random process is also stationary in the strict sense.

Multiple Random processes

for two real random processes $x(t)$ and $y(t)$, we define the cross-correlation function

$$R_{xy}(t_1, t_2) = \overline{x(t_1) y(t_2)}$$

The two processes are said to be jointly stationary if each of the processes is individually wide-sense stationary and if

$$R_{xy}(t_1, t_2) = R_{xy}(t_2 - t_1) = R_{xy}(\tau)$$

1) Uncorrelated, orthogonal and Independent processes -

Two processes $x(t)$ and $y(t)$ are said to be uncorrelated if their cross-correlation function is equal to the product of their means, that is,

$$R_{xy}(\tau) = \overline{x(t) y(t+\tau)} = \bar{x} \bar{y}$$

$x(t)$ and $y(t+\tau)$ are uncorrelated for all t and τ .

Processes $x(t)$ and $y(t)$ are said to be orthogonal if

$$R_{xy}(\tau) = 0$$

Processes $x(t)$ and $y(t)$ are independent random processes if the random variables $x(t_1)$ and $y(t_2)$ are independent for all possible choices of t_1 and t_2 .

Sum of Random processes

(14)

If two stationary processes $x(t)$ and $y(t)$ are added to form a process $z(t)$, the statistics of $z(t)$ can be determined in terms of those of $x(t)$ and $y(t)$.

$$\text{If } z(t) = x(t) + y(t)$$

then

$$R_z(z) = \overline{z(t) z(t+z)} = \overline{[x(t) + y(t)] [x(t+z) + y(t+z)]}$$

$$= R_x(z) + R_y(z) + R_{xy}(z) + R_{yx}(z)$$

If $x(t)$ and $y(t)$ are uncorrelated, then

$$R_{xy}(z) = R_{yx}(z) = \bar{x} \bar{y}$$

$$\text{and } R_z(z) = R_x(z) + R_y(z) + 2 \bar{x} \bar{y}$$

Most processes of interest in communication problems have zero means. If processes $x(t)$ and $y(t)$ are uncorrelated with either $\bar{x}$ or $\bar{y} = 0$, then

$$R_z(z) = R_x(z) + R_y(z)$$

$$\text{and } S_z(f) = S_x(f) + S_y(f)$$

$$\overline{z^2} = \overline{x^2} + \overline{y^2}$$

Mean square of a sum of orthogonal or incoherent processes is equal to the sum of the mean squares of these processes.

Multiplication of Random Processes

Consider the case when two independent, jointly wide-sense stationary random processes are multiplied together,

$$W(t) = X(t)Y(t)$$

The mean of $W(t)$ is

$$E[W(t)] = \mu_W = E[X(t)] E[Y(t)] = \mu_X \mu_Y$$

Variance of $W(t)$ is

$$\sigma_W^2 = E[\{X(t)Y(t) - \mu_X \mu_Y\}^2]$$

$$= (\sigma_X^2 + \mu_X^2)(\sigma_Y^2 + \mu_Y^2) - \mu_X^2 \mu_Y^2$$

The autocorrelation for $W(t)$ is

$$\begin{aligned} R_W(\tau) &= E[X(t)Y(t)X(t+\tau)Y(t+\tau)] \\ &= E[X(t)X(t+\tau)] E[Y(t)Y(t+\tau)] \\ &= R_X(\tau) R_Y(\tau) \end{aligned}$$

We see a wide sense stationary situation in which the autocorrelation of a product is the product of the autocorrelations.

Ex

- 1) Consider the sinusoidal random process $x(t) = A \sin(\omega_0 t)$ and A is a uniform RV over $[0, 1]$. Find Mean & Autocorrelation function. Check if the process is WSS?

Sol :-

$$\mu_x(t) = E[x(t)] = E[A \sin(\omega_0 t)] = E[A] \sin \omega_0 t$$

$$E[A] = \int_{-\infty}^{\infty} A P_X(A) dA = \int_0^1 A dA = \frac{A^2}{2} \Big|_0^1 = \frac{1}{2}$$

$$\mu_x(t) = \frac{1}{2} \sin(\omega_0 t)$$

Not stationary

$\therefore t = 0, \pm \pi, \mu = 0$

$$\omega_0 t = \pi/2, \mu_x(t) = 1/2$$

$$R_{xx}(t_1, t_2) = E[x(t_1) x(t_2)]$$

$$= E[A^2 \sin(\omega_0 t_1) \sin(\omega_0 t_2)] = \frac{A^3}{3} \Big|_0^1 \cdot \sin(\omega_0 t_1) \sin(\omega_0 t_2)$$

$$R_{xx}(t, t+\tau) = \frac{1}{3} \sin(\omega_0 t) \sin(\omega_0 (t+\tau))$$

ACF depends on t_1 and t_2

- 2) Consider a sinusoid with a f uniformly distributed over some interval $(0, f_0)$. $x(t) = \cos(2\pi f t)$, $f \rightarrow$ RV. Find mean function.

Sol :- $\mu_x(t) = E[\cos(2\pi f t)] = \frac{1}{f_0} \int_0^{f_0} \cos(2\pi f t) df = \frac{\sin(2\pi f_0 t)}{2\pi f_0 t}$

$$= \text{sinc}(2f_0 t)$$

Ex ① Show that random process

$X(t) = A \cos(\omega_c t + \theta)$ is a wide-sense stationary process. where $\theta \rightarrow$ RV distributed in the range $(0, 2\pi)$.

Soln - For any sample function, the phase is equally likely to have any value in the range $(0, 2\pi)$.

$$E[X(t)] = E[A \cos(\omega_c t + \theta)] = AE[\cos(\omega_c t + \theta)] \\ = A \int_0^{2\pi} \cos(\omega_c t + \theta) P_\theta(\theta) d\theta$$

$$E[X(t)] = \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega_c t + \theta) d\theta = 0 \rightarrow \textcircled{1}$$

Thus, the ensemble mean of sample function amplitudes at any instant t is zero.

The autocorrelation function $R_X(t_1, t_2)$ is

$$R_X(t_1, t_2) = E[X(t_1) X(t_2)] = A^2 E[\cos(\omega_c t_1 + \theta) \cos(\omega_c t_2 + \theta)] \\ = \frac{A^2}{2} E[\underbrace{\cos[\omega_c(t_2 - t_1)]}_{\substack{\uparrow \\ \text{Contains no RV}}} + \underbrace{\cos[\omega_c(t_2 + t_1) + 2\theta]}_{\substack{\uparrow \\ \text{leads to zero}}}]$$

$$R_X(t_1, t_2) = \frac{A^2}{2} \cos[\omega_c(t_2 - t_1)]$$

$$R_X(\tau) = \frac{A^2}{2} \cos(\omega_c \tau), \quad \tau = t_2 - t_1 \rightarrow \textcircled{2}$$

From ① & ② it is clear that $X(t)$ is a wide-sense stationary process.

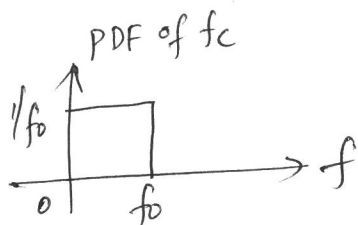
Ex

① Consider $x(t) = A_c \cos(2\pi f_c t + \phi)$

$f_c \rightarrow$ RV uniformly distributed from 0 to f_0 .

Find the Mean.

Sol:-



$\phi = 0$.

$$\begin{aligned} \mu_x(t) &= E[A_c \cos(2\pi f_c t + \phi)] \\ &= A_c \int_0^{f_0} \cos(2\pi f_c t + \phi) \frac{1}{f_0} df = \frac{\sin 2\pi f_0 t}{2\pi f_0 t} \end{aligned}$$

$$\mu_x(t) = \text{Sinc}(2f_0 t)$$

② Let $x(t)$ and $y(t)$ are zero-mean & WSS random processes.
Consider the random process $z(t) = x(t) + y(t)$

a) Determine the ACF and power spectral density of $z(t)$ if $x(t)$ and $y(t)$ are jointly WSS.

b) Repeat part (a) if $x(t)$ and $y(t)$ are orthogonal.

c) If $x(t)$ and $y(t)$ are orthogonal, the mean square value of $z(t)$?

Sol:-

$$\begin{aligned} \text{a) } R_{zz}(t_1, t_2) &= E[z(t_1)z(t_2)] \\ &= E[(x(t_1) + y(t_1))(x(t_2) + y(t_2))] \\ &= E[x(t_1)x(t_2) + y(t_2)x(t_1) + y(t_1)x(t_2) + y(t_1)y(t_2)] \\ &= R_{xx}(t_1, t_2) + R_{xy}(t_1, t_2) + R_{yx}(t_1, t_2) + R_{yy}(t_1, t_2) \end{aligned}$$

Ex

(6)

Consider the Rf $X(t) = A \cos(\omega t) + B \sin(\omega t)$, where A and B are independent, zero-mean Gaussian random variables with equal variances of σ^2 . This random process is formed as a linear combination of two Gaussian RV's and therefore the samples of this process are also Gaussian random variables. Find the mean & autocorrelation of the process.

Soln:-

$$\mu_X(t) = E[A \cos(\omega t) + B \sin(\omega t)]$$

$$= E[A] \cos(\omega t) + E[B] \sin(\omega t) = 0$$

$$R_{XX}(t_1, t_2) = E[(A \cos(\omega t_1) + B \sin(\omega t_1))(A \cos(\omega t_2) + B \sin(\omega t_2))]$$

$$= E[A^2] \cos(\omega t_1) \cos(\omega t_2) + E[B^2] \sin(\omega t_1) \sin(\omega t_2)$$

$$+ E[AB] \{ \cos(\omega t_1) \sin(\omega t_2) + \sin(\omega t_1) \cos(\omega t_2) \}$$

$$= \sigma^2 \{ \cos(\omega t_1) \cos(\omega t_2) + \sin(\omega t_1) \sin(\omega t_2) \}$$

$$= \sigma^2 \cos(\omega(t_2 - t_1)) = \sigma^2 \cos(\omega \tau)$$

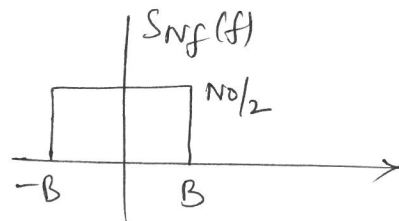
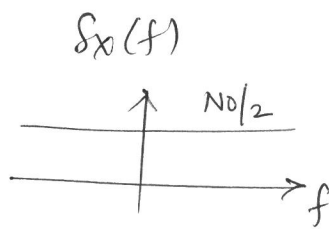
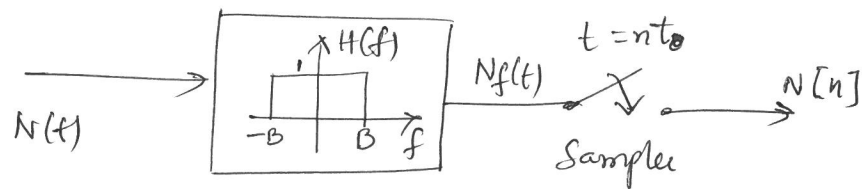
Note that this process is WSS since the mean is constant & the autocorrelation function depends only on the time difference.

Since the process is zero-mean, the first order PDF is that of a zero-mean Gaussian RV.

$$f_X(x, t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

This PDF is independent of time as would be expected for a stationary random process.

is convert a white noise process from Continuous time to discrete time using a sampler. Since white noise has infinite power, it cannot be sampled directly and must be filtered first. we use a ideal LPF of BW 'B' to perform the sampling so that



Let $N_f(t)$ be the random process at the output of the LPF. This process has a PSD of

$$S_{N_f N_f}(f) = \frac{N_0}{2} \text{rect}\left(\frac{f}{2B}\right) = \begin{cases} \frac{N_0}{2}, & |f| < B \\ 0, & \text{otherwise} \end{cases}$$

The corresponding autocorrelation function is,

$$R_{N_f N_f}(\tau) = N_0 B \text{Sinc}(2B\tau)$$

Ex: - Determine the PSD & mean square value of a random process (11)
 $x(t) = A \cos(\omega_c t + \theta)$

where $\theta \rightarrow$ random variable uniformly distributed over $(0, 2\pi)$

Soln - $R_x(\tau) = \frac{A^2}{2} \cos \omega_c \tau$

$$S_x(f) = \frac{A^2}{4} [\delta(f + f_c) + \delta(f - f_c)]$$

$$P_x = \overline{x^2} = R_x(0) = \frac{A^2}{2}$$

Ex: Determine the autocorrelation function and the PSD of the DSB-SC modulated process $m(t) \cos(\omega_c t + \theta)$, where $m(t)$ is a wide-sense stationary random process and θ is a RV uniformly distributed over $(0, 2\pi)$ and independent of $m(t)$.

Soln - Let $\phi(t) = m(t) \cos(\omega_c t + \theta)$

$$R_\phi(t) = \overline{m(t) \cos(\omega_c t + \theta) \cdot m(t+\tau) \cos(\omega_c(t+\tau) + \theta)}$$

Because $m(t)$ and θ are independent

$$\begin{aligned} R_\phi(t) &= \overline{m(t) m(t+\tau) \cos(\omega_c t + \theta) \cos[\omega_c(t+\tau) + \theta]} \\ &= \frac{1}{2} R_m(\tau) \cos(\omega_c \tau) \end{aligned}$$

$$S_\phi(f) = \frac{1}{4} [S_m(f + f_c) + S_m(f - f_c)]$$

$$\overline{\phi^2(t)} = R_\phi(0) = \frac{1}{2} R_m(0) = \frac{1}{2} \overline{m^2(t)}$$

Hence, the power of the DSB-SC modulated signal is half the power of the modulating signal.